



IBM: Applied Data Science Capostone

YOUSSEF GHOREBL
19 APRIL, 2026

GitHub url :<https://github.com/youssefgh10/Applied-Data-Science-capstone>

image: <https://qz.com/cdn-cgi/image/width=1920,quality=85,format=auto/https://assets.qz.com/media/GettyImages-2158701295.jpg>

OUTLINE

1 Executive Summary

2 Introduction

3 Methodology

4 Results

5 Discussion & Conclusion

EXECUTIVE SUMMARY

Goal: Predict whether the Falcon 9 first stage will land successfully, enabling accurate launch cost estimation.

SpaceX lists Falcon 9 launches at ~\$62M vs. \$165M+ for competitors , savings driven by stage one reuse.

Key steps: data collection via API & web scraping → EDA → interactive visualization → ML classification.

Several features (orbit type, payload mass, flight number) show correlation with landing outcome.

Decision Tree achieved the highest cross-validation score among four ML models tested.

INTRODUCTION

The Business Problem

This capstone project aims to predict whether the first stage of a Falcon 9 rocket will land successfully. SpaceX promotes its launches at a cost of about \$62 million, while other providers often charge more than \$165 million per launch. A large part of this cost advantage comes from the ability to recover and reuse the first stage.

By estimating the likelihood of a successful landing, we can better assess the true cost of a launch. Such insights could help competing companies make more informed and competitive bids when challenging SpaceX in the commercial launch market.

Research Question

Given a set of features about a Falcon 9 launch like (payload mass, orbit type, launch site, booster version, flight number, and more), will the first stage land successfully?

Key Note

Not all unsuccessful landings are failures , SpaceX sometimes performs intentional controlled ocean landings. Class 0 = did not land (any reason). Class 1 = successful landing.

METHODOLOGY

This is a 4-phase pipeline which includes:

1 Data Collection & Wrangling

SpaceX REST API
Wikipedia web scraping
Missing value imputation (mean)
One-hot encoding

2 Exploratory Data Analysis

Pandas & NumPy for summary stats
SQL queries on launch database

3 Data Visualization

Matplotlib & Seaborn (static charts)
Folium (interactive maps)
Plotly Dash (interactive dashboard)

4 Machine Learning Prediction

Logistic Regression
SVM
Decision Tree
KNN
GridSearchCV tuning

METHODOLOGY

① Data Collection & Wrangling

SpaceX REST API

- Endpoint:** `api.spacexdata.com/v4/rockets/`
- Filtered to Falcon 9 launches only
 - Missing values replaced with column mean
 - Result: 90 rows × 17 columns

FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs		LandingPad	Block	ReusedCount	Serial	Longitude	Latitude
1	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None	1	False	False	False		None	1.0	0	B0003	-80.577366	28.5618
2	2012-05-22	Falcon 9	525.0	LEO	CCSFS SLC 40	None None	1	False	False	False		None	1.0	0	B0005	-80.577366	28.5618
3	2013-03-01	Falcon 9	677.0	ISS	CCSFS SLC 40	None None	1	False	False	False		None	1.0	0	B0007	-80.577366	28.5618
4	2013-09-29	Falcon 9	500.0	PO	VAFB SLC 4E	False Ocean	1	False	False	False		None	1.0	0	B1003	-120.610829	34.6320
5	2013-12-03	Falcon 9	3170.0	GTO	CCSFS SLC 40	None None	1	False	False	False		None	1.0	0	B1004	-80.577366	28.5618

github:<https://github.com/youssefgh10/Applied-Data-Science-capstone/blob/main/1-Data%20Collection.ipynb>

METHODOLOGY

① Data Collection & Wrangling

Web Scraping (Wikipedia)

- Scraped Falcon 9 & Heavy launch table
- Falcon 9 launches only retained
- Result: 121 rows × 11 columns

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success\n	F9 v1.0B0003.1	Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.0B0004.1	Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.0B0005.1	No attempt\n	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success\n	F9 v1.0B0006.1	No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success\n	F9 v1.0B0007.1	No attempt\n	1 March 2013	15:10

- The dataset is then cleaned to remove any missing values, and categorical variables are transformed using one-hot encoding.
- An additional column named “Class” is introduced, where a value of 0 represents a failed launch and 1 indicates a successful one.
- After preprocessing, the final dataset consists of 90 observations (rows) and 83 features (columns).

github url:

<https://github.com/youssefgh10/Applied-Data-Science-capstone/blob/main/2-Data%20Collection%20using%20Web%20Scraping.ipynb>

<https://github.com/youssefgh10/Applied-Data-Science-capstone/blob/main/3-Data%20Wrangling.ipynb>

METHODOLOGY

② Exploratory Data Analysis

SQL Queries

Also the data is queried with SQL to know more about:

- Unique launch site names
- Total payload mass by NASA (CRS)
- Avg payload mass by booster F9 v1.1
- Date of first ground-pad success
- Drone-ship failures & landing outcome counts

Votre texte de paragraphe

Pandas & NumPy

Functions from the Pandas and NumPy libraries are utilized to extract essential insights from the collected dataset, including:

- Launches per launch site
- Occurrence of each orbit type
- Mission outcome counts
- Feature correlations with landing class

github url:

<https://github.com/youssefgh10/Applied-Data-Science-capstone/blob/main/4-EDA%20with%20SQL.ipynb>

METHODOLOGY

3

Data Visualization

Visualization Tools

1-Matplotlib and Seaborn

- The Matplotlib and Seaborn libraries are used to create visual representations of the data, including scatter plots, bar charts, and line graphs. These visualizations help reveal patterns and relationships between different variables, such as:
- How flight number varies across launch sites
- The connection between payload mass and launch site
- the relationship between orbit type and the success rate

2-Folium

- The Folium library is applied to build interactive maps for geographic analysis. It is used to:
- Display all launch sites on a map
- Highlight successful and failed launches at each location
- Show distances between launch sites and nearby points of interest, such as cities, railways, and highways

3-Dash

The Dash framework is used to build an interactive web application that allows users to explore the data through controls such as dropdown menus and range sliders.

With the help of visualizations like pie charts and scatter plots, the app enables users to:

- View the total number of successful launches for each launch site
- Examine the relationship between payload mass and mission outcome (success or failure) across different launch sites

<https://github.com/youssefgh10/Applied-Data-Science-capstone/blob/main/5-EDA%20with%20Visualization.ipynb>

<https://github.com/youssefgh10/Applied-Data-Science-capstone/blob/main/6-Interactive%20Visual%20Analytics%20with%20Folium.ipynb>

<https://github.com/youssefgh10/Applied-Data-Science-capstone/blob/main/7-Interactive%20Visual%20Analytics%20with%20Dashboard.py>

METHODOLOGY

4 Machine Learning Prediction

ML Models

Functions from the Scikit-learn library are used to develop and evaluate the machine learning models.

The prediction phase involves several key steps:

- Dividing the unscaled dataset into training and testing subsets
- Fitting scaling inside each model pipeline and validation fold
- Building different classification models, including:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Tree
 - K-Nearest Neighbors (KNN)
- Training each model using the training data
- Tuning hyperparameters to identify the optimal configuration for each algorithm
- Assessing model performance using metrics such as accuracy and the confusion matrix

<https://github.com/youssefgh10/Applied-Data-Science-capstone/blob/main/8-Machine%20learning%20prediction.ipynb>

RESULTS

The results are organized into five main sections:

- SQL (exploratory data analysis using SQL)
- Matplotlib and Seaborn (visual exploratory analysis)
- Folium
- Dash
- Predictive Analysis

In all the visualizations presented, a value of 0 corresponds to a failed launch, while a value of 1 indicates a successful launch.

RESULTS

① EDA with SQL

Key findings from SQL queries on the launch database:

Unique launch sites

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

5 records where launch sites begin with ‘CCA’

Launch_Site
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40

Total payload mass carried by boosters launched by NASA (CRS)

Total_payloadmass_By_NASA_CRS
45596

The average payload mass carried by booster version F9 v1.1

Avg_payload_BoosterVF9_V1.1
2928.4

The date when the first successful landing outcome in ground pad was achieved

Date_firstsuccessfullanding_groundpad
2015-12-22

The names of the booster versions which have carried the maximum payload mass

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

The names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

The failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

Month	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

The total number of successful and failure mission outcomes

Outcome	total
Failure	1
Success	100

The count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order

Landing_Outcome	CNT
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

RESULTS

① EDA with SQL

Key findings from SQL queries on the launch database:

Unique launch sites

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

5 records where launch sites begin with ‘CCA’

Launch_Site
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40

Total payload mass carried by boosters launched by NASA (CRS)

Total_payloadmass_By_NASA_CRS
45596

The average payload mass carried by booster version F9 v1.1

Avg_payload_BoosterVF9_V1.1
2928.4

The date when the first successful landing outcome in ground pad was achieved

Date_firstsuccessfullanding_groundpad
2015-12-22

The names of the booster versions which have carried the maximum payload mass

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

The names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

The failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

Month	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

The total number of successful and failure mission outcomes

Outcome	total
Failure	1
Success	100

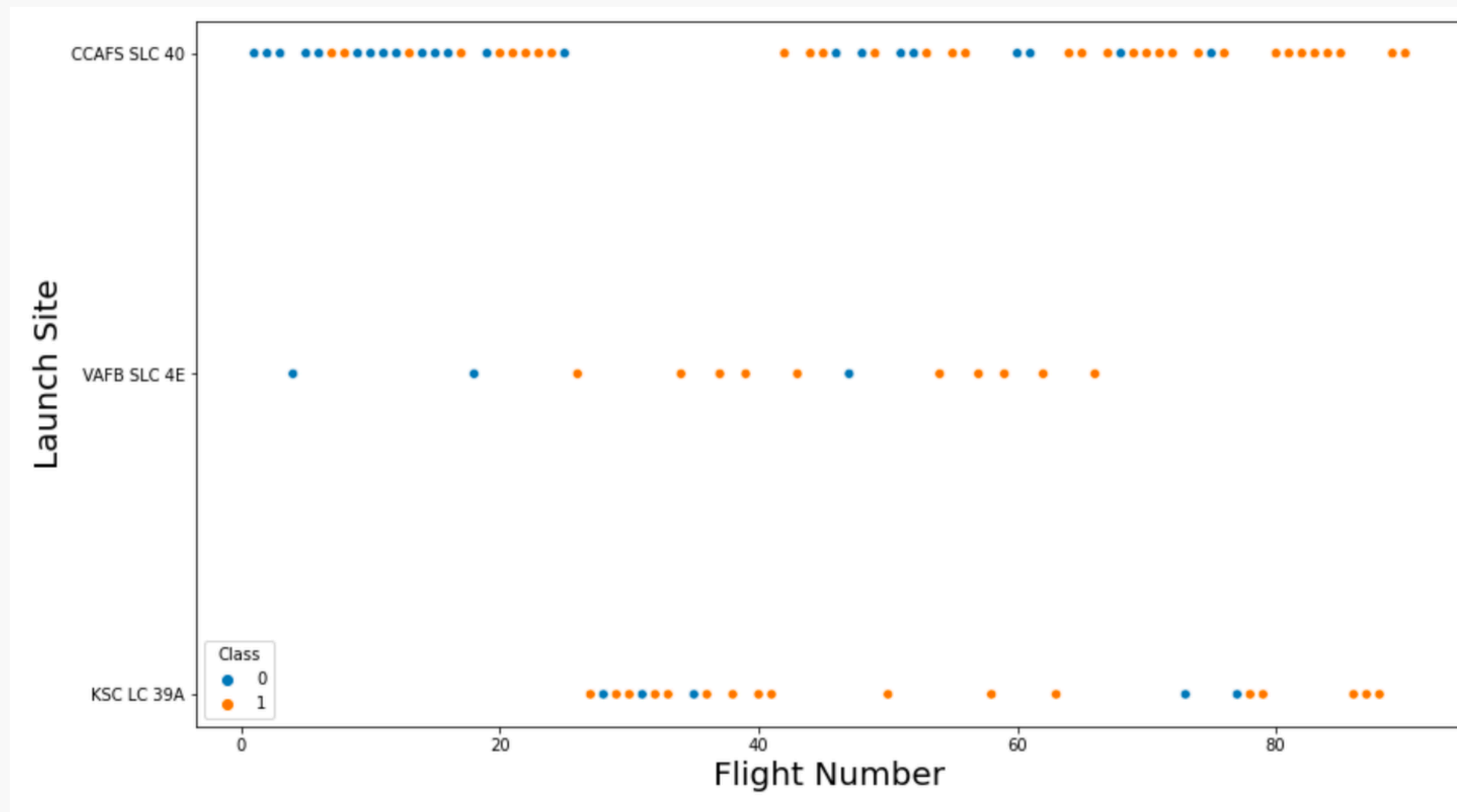
The count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order

Landing_Outcome	CNT
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

RESULTS

② EDA with Visualization (Matplotlib & Seaborn)

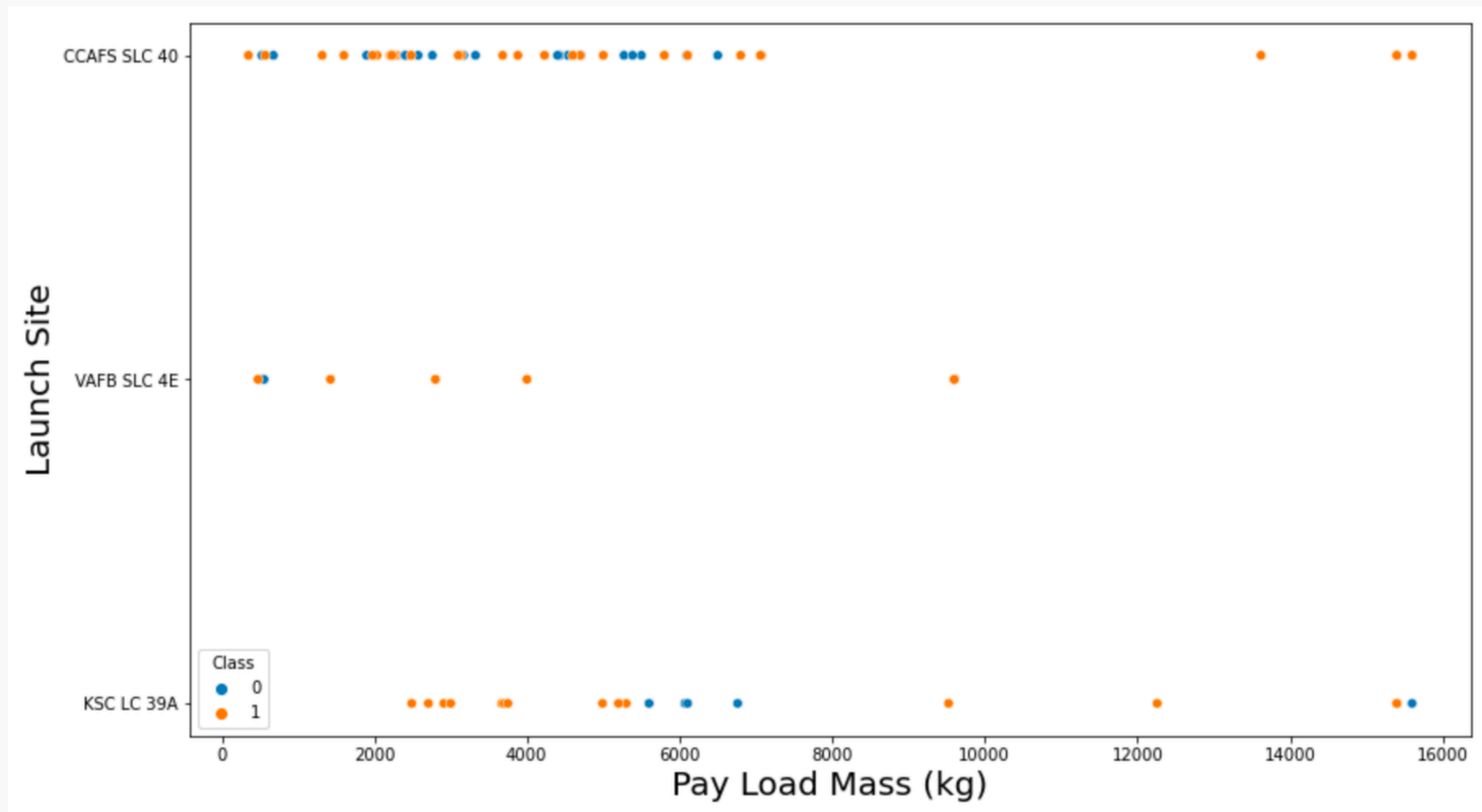
The relationship between flight number and launch site



RESULTS

② EDA with Visualization (Matplotlib & Seaborn)

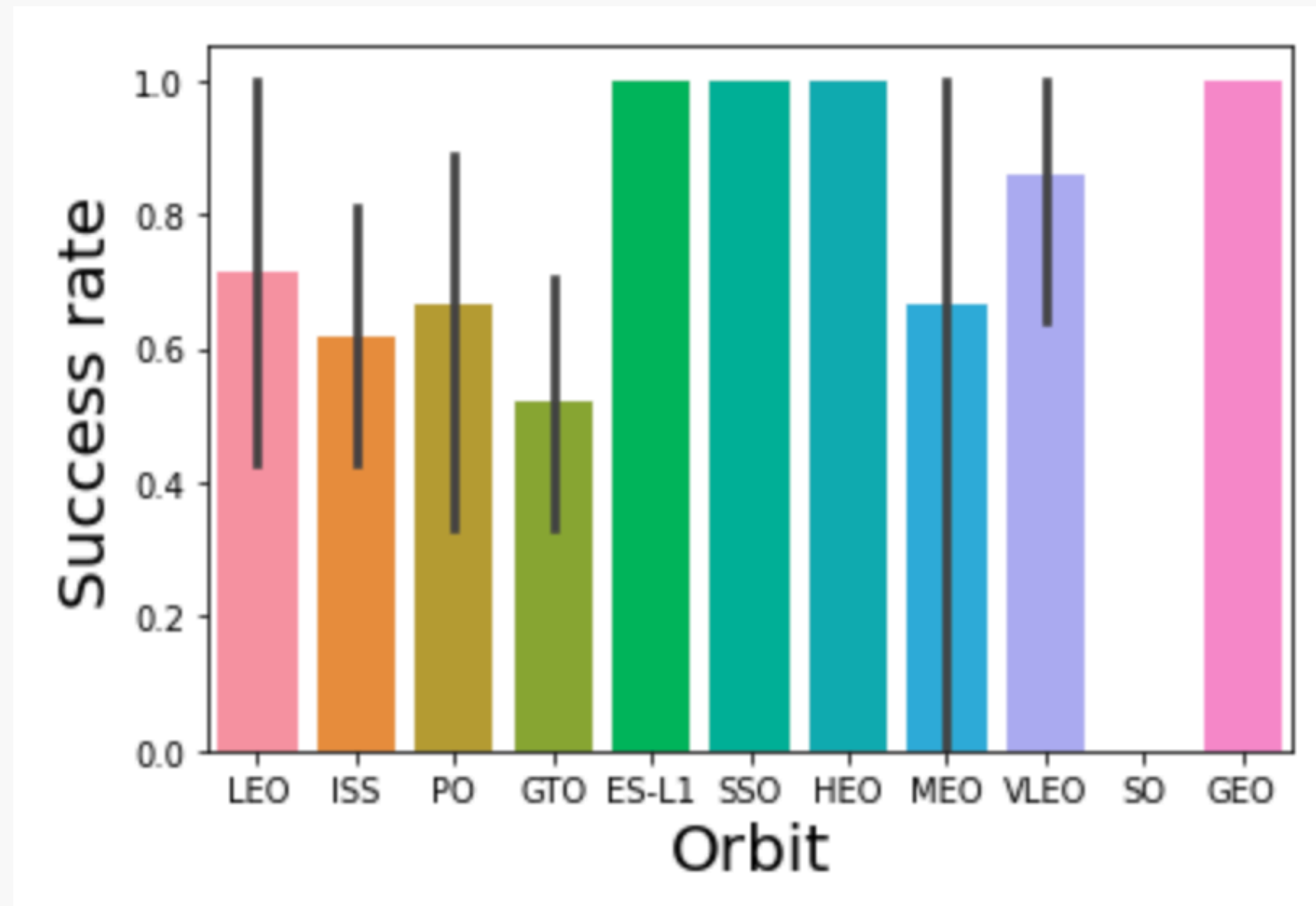
The relationship between payload mass and launch site



RESULTS

② EDA with Visualization (Matplotlib & Seaborn)

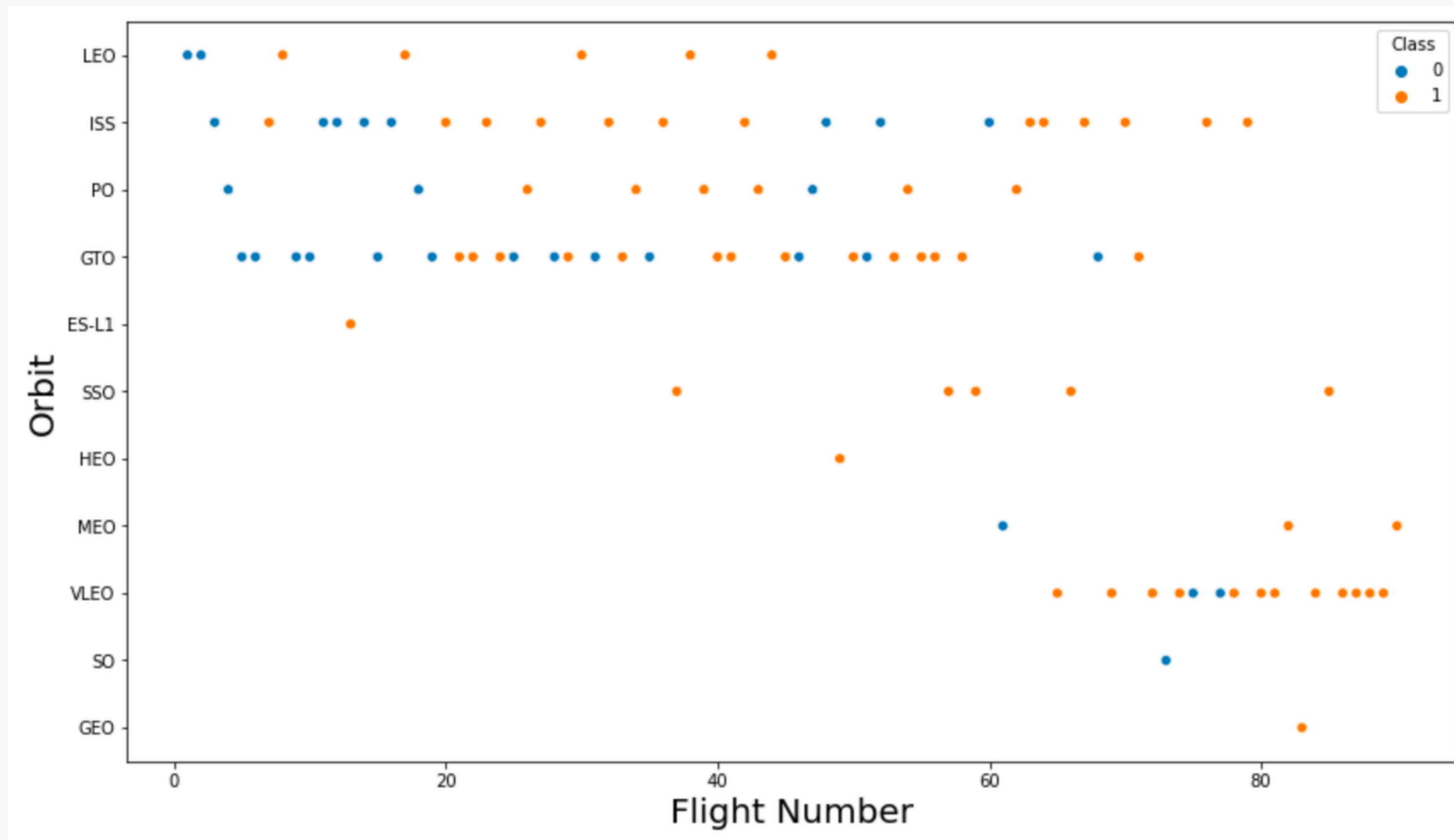
The relationship between success rate and orbit type



RESULTS

② EDA with Visualization (Matplotlib & Seaborn)

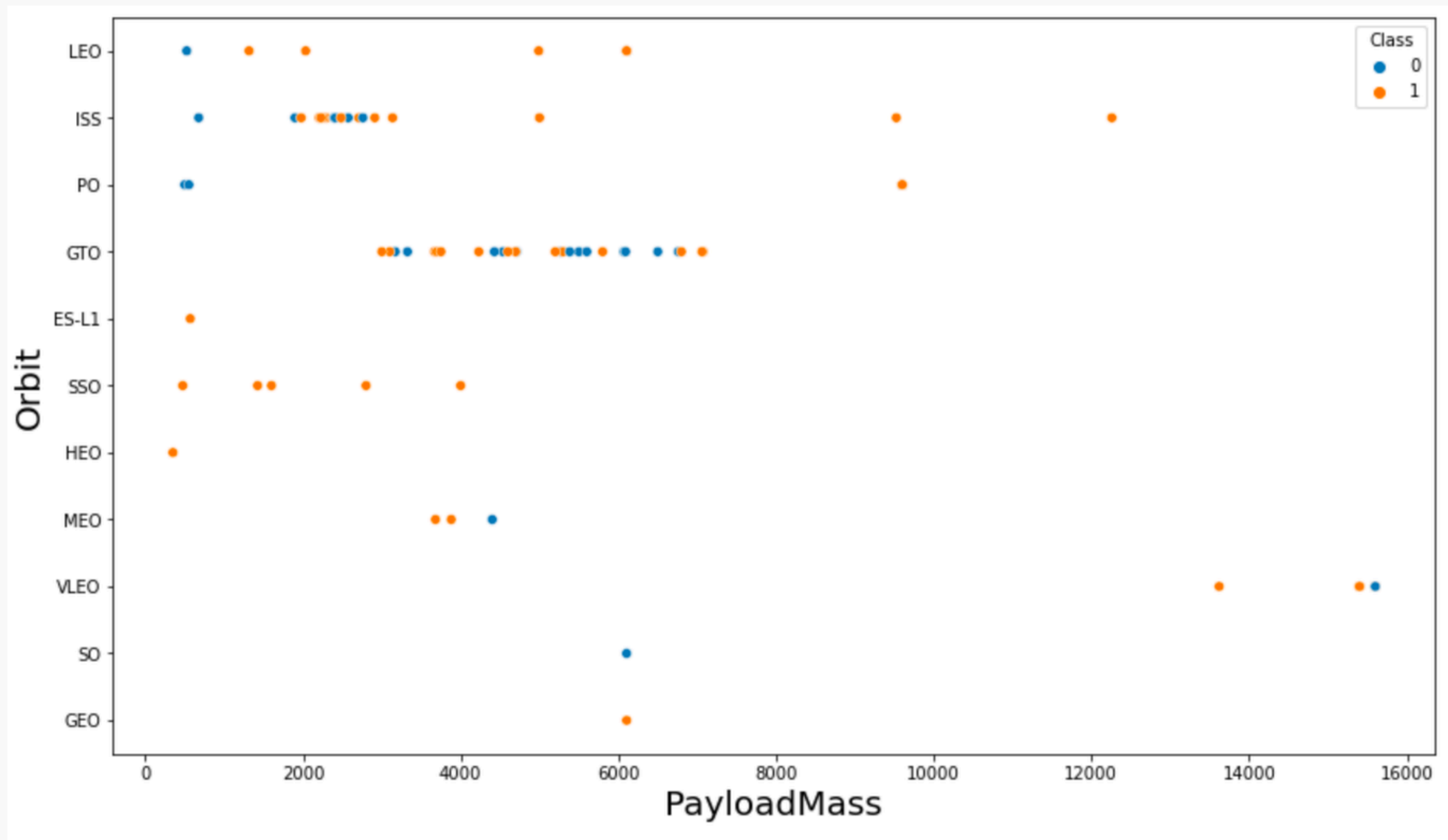
The relationship between flight number and orbit type



RESULTS

② EDA with Visualization (Matplotlib & Seaborn)

The relationship between payload mass and orbit type

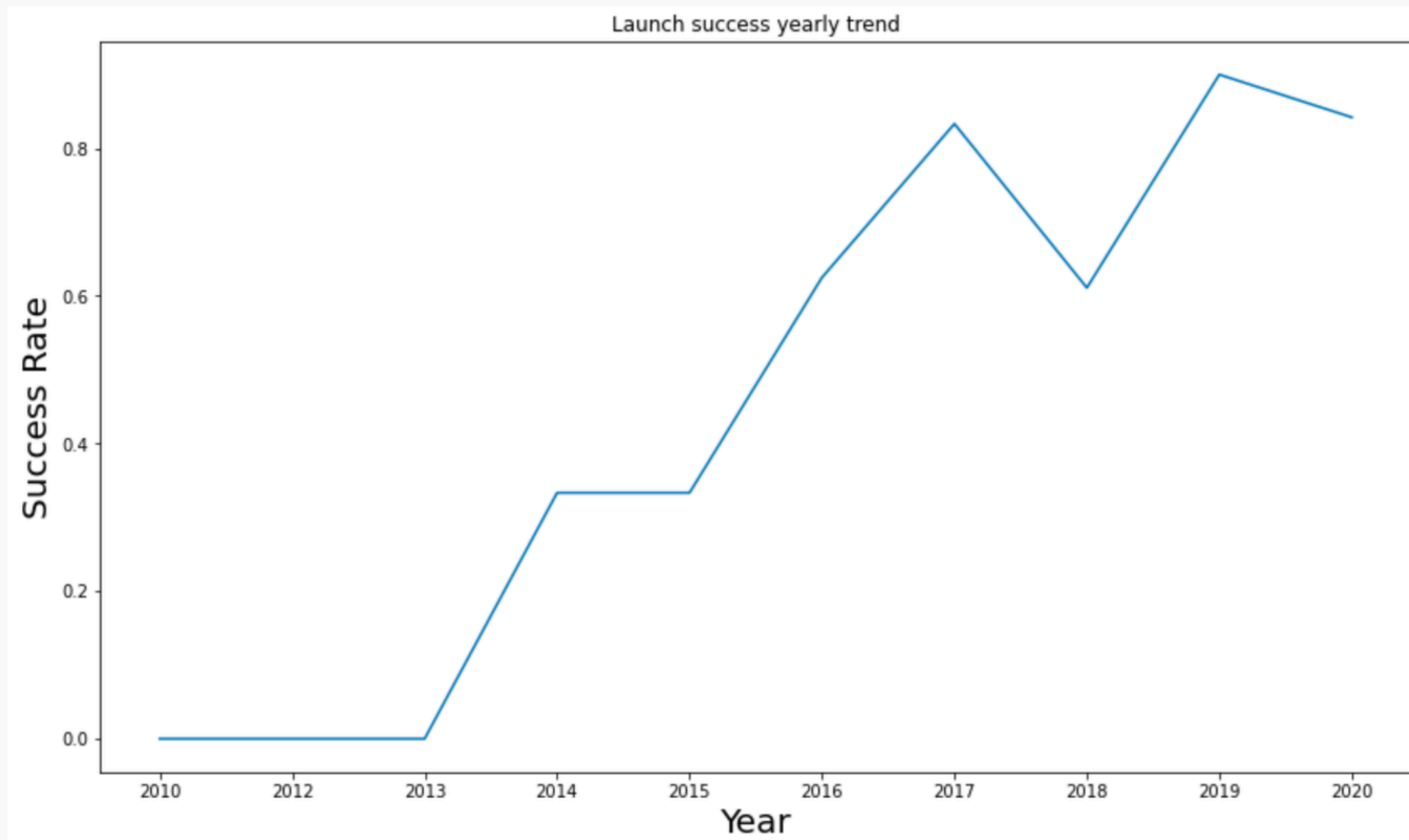


RESULTS

② EDA with Visualization (Matplotlib & Seaborn)

Key insight: Higher flight numbers correlate with more successful landings so SpaceX improved over time.

The launch success yearly trend

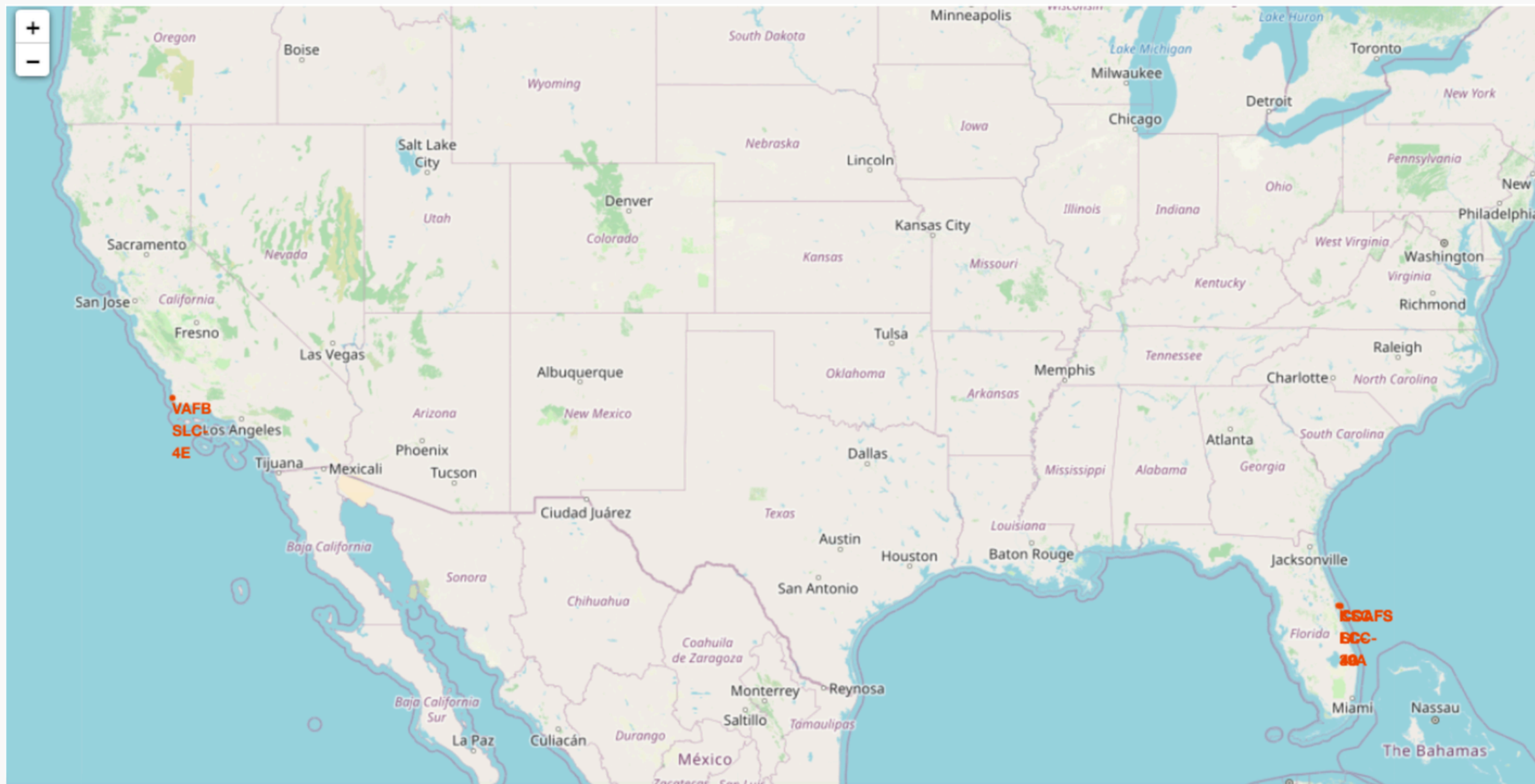


RESULTS

③ Interactive Map Folium

Key insight: All launch sites are coastal, minimizing risk from failed launches over populated areas.

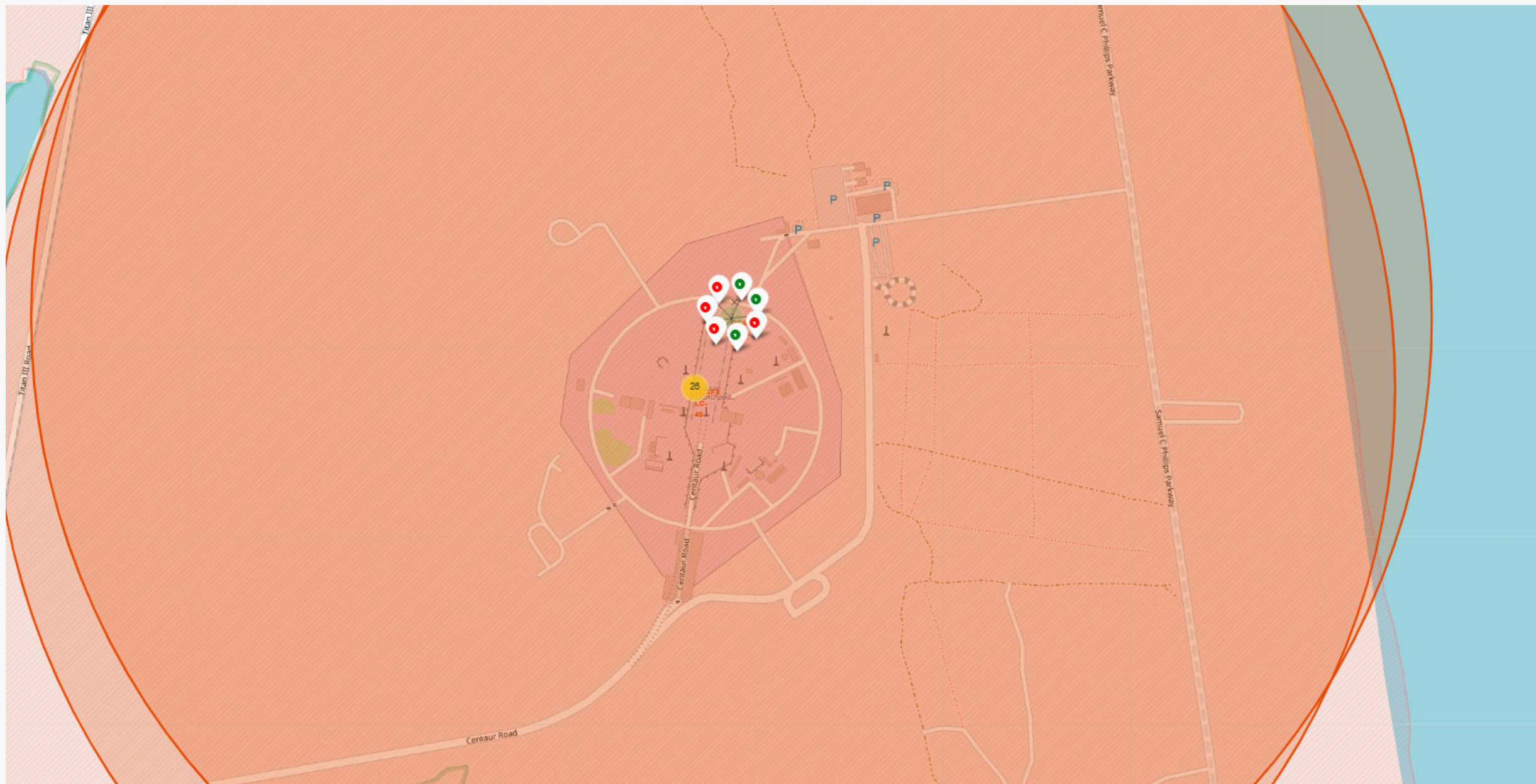
All launch sites on map



RESULTS

③ Interactive Map Folium

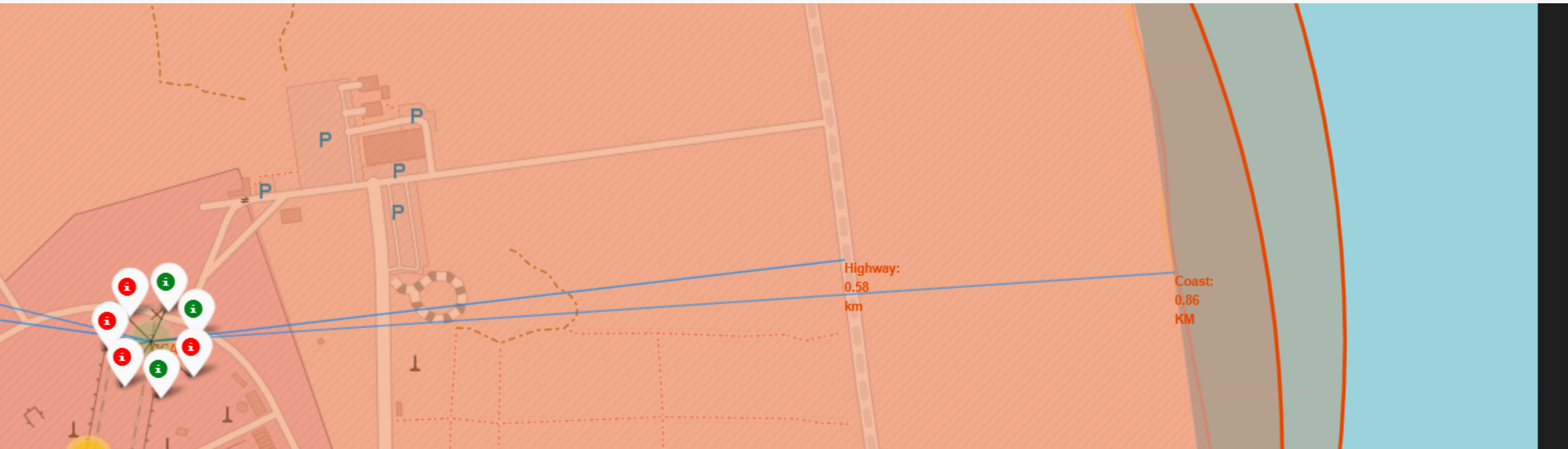
- The succeeded launches and failed launches for each site on map
 - If we zoom in on one of the launch site, we can see green and red tags. Each green tag represents a successful launch while each red tag represents a failed launch



RESULTS

③ Interactive Map — Folium

- The distances between a launch site to its proximities such as the nearest city, railway, or highway
 - The picture below shows the distance between the CCAFS SLC-40 launch site and the nearest coastline and also nearest Highway



RESULTS

④ Interactive Dashboard :Plotly Dash

- The picture below shows a pie where All sites are chosen.
- It represents the Total Success of each Site

Total Success Launches by All Sites



RESULTS

④ Interactive Dashboard :Plotly Dash

- The picture below shows a pie chart when launch site KSC LC-39A is chosen.
- 0 represents failed launches while 1 represents successful launches. We can see that 76.9% of launches done are successful launches.

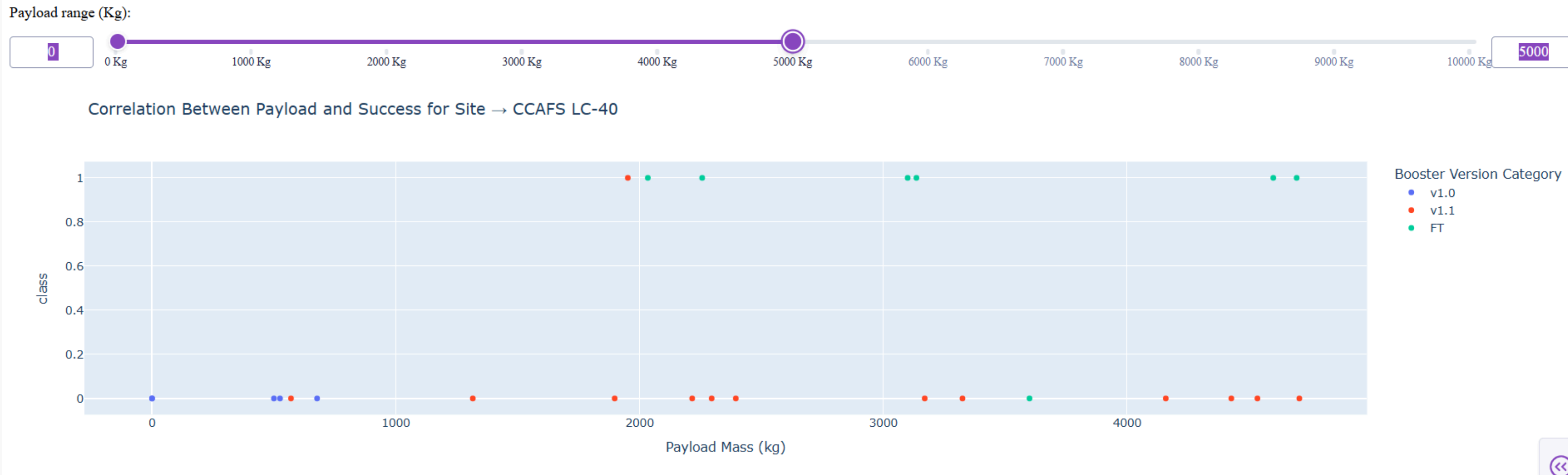
Total Success Launches for Site → KSC LC-39A



RESULTS

④ Interactive Dashboard :Plotly Dash

- The picture below shows a scatterplot when the payload mass range is set to be from 0kg to 5000kg for Site CCAFS LC-40.
- Class 0 represents failed launches while class 1 represents successful launches.



RESULTS

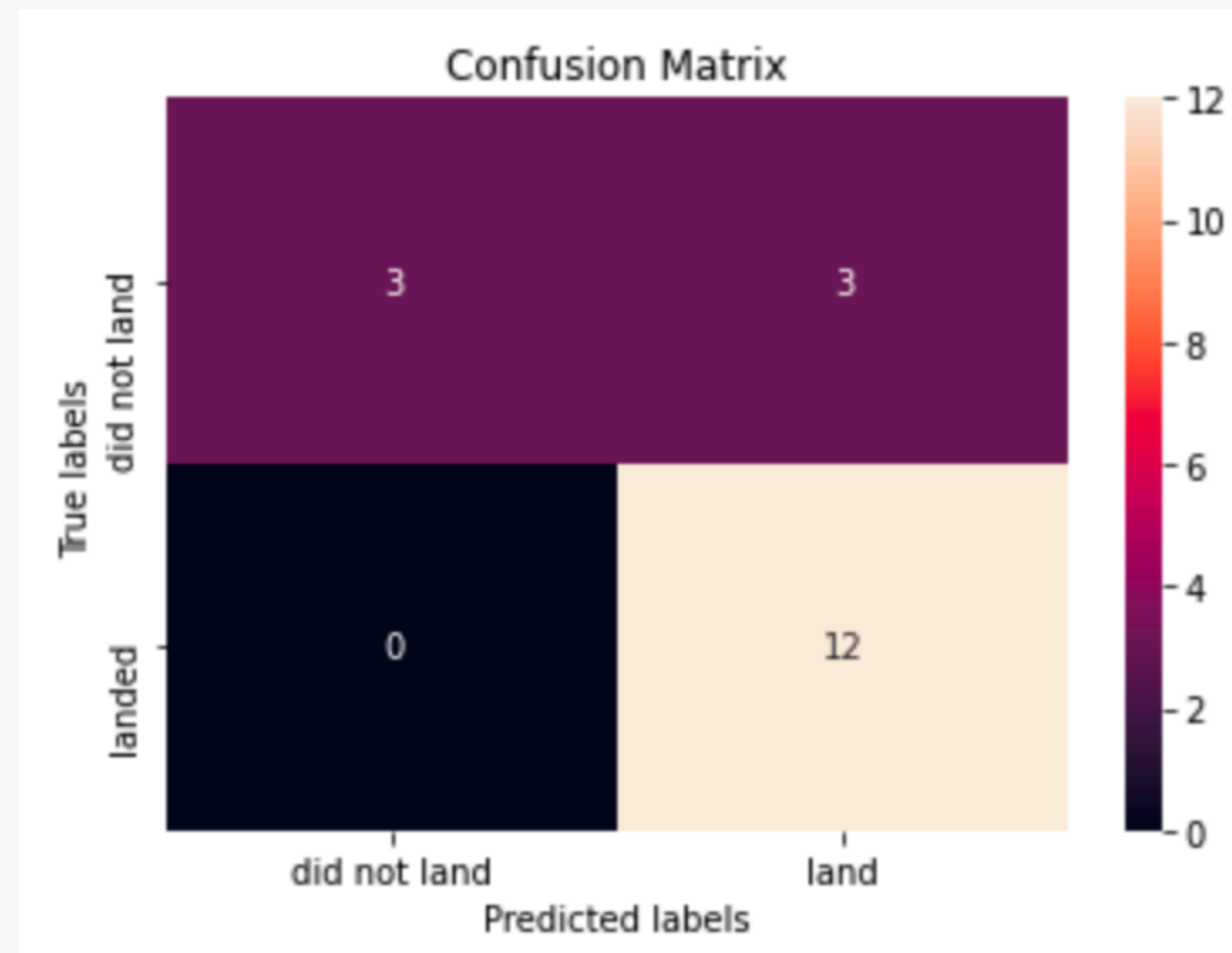
⑤ Predictive Analysis : Classification Models

- Logistic regression

GridSearchCV best score: 0.8464285714285713

Accuracy score on test set: 0.8333333333333333

Confusion matrix:



RESULTS

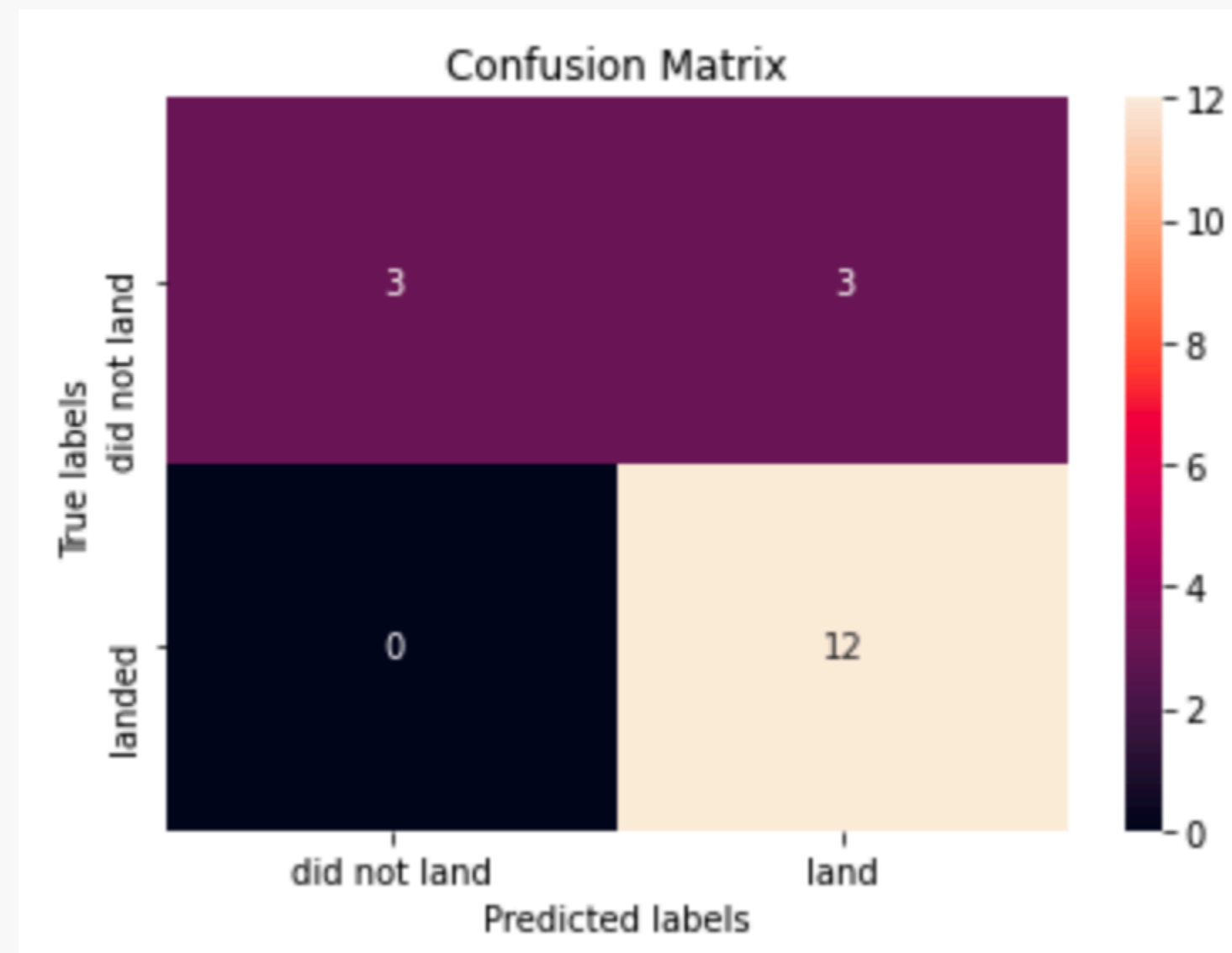
⑤ Predictive Analysis : Classification Models

- Support vector machine (SVM)

GridSearchCV best score: 0.8214285714285714

Accuracy score on test set: 0.8333333333333333

Confusion matrix:



RESULTS

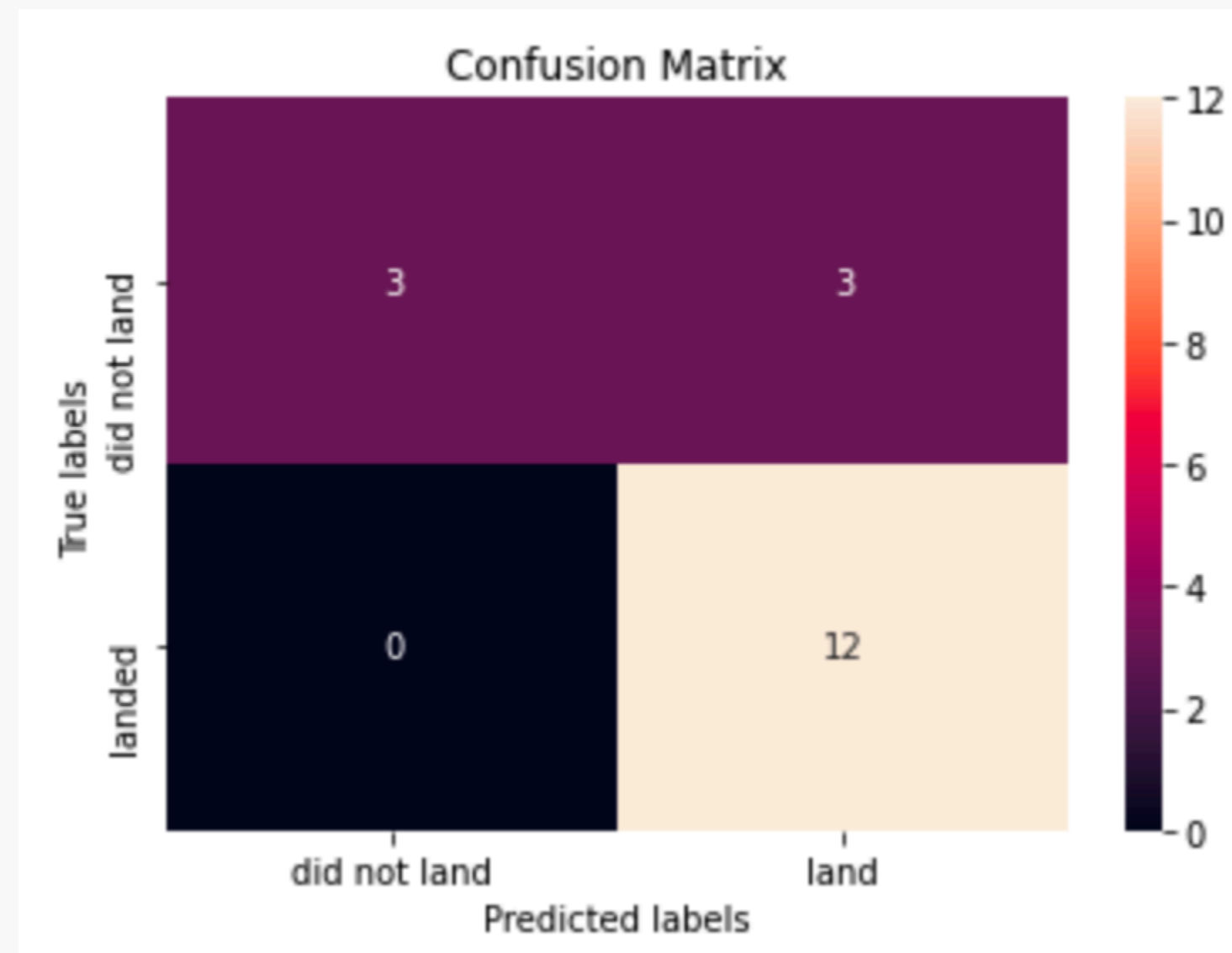
⑤ Predictive Analysis : Classification Models

- Decision tree

GridSearchCV best score: 0.8767857142857143

Accuracy score on test set: 0.8333333333333333

Confusion matrix:



RESULTS

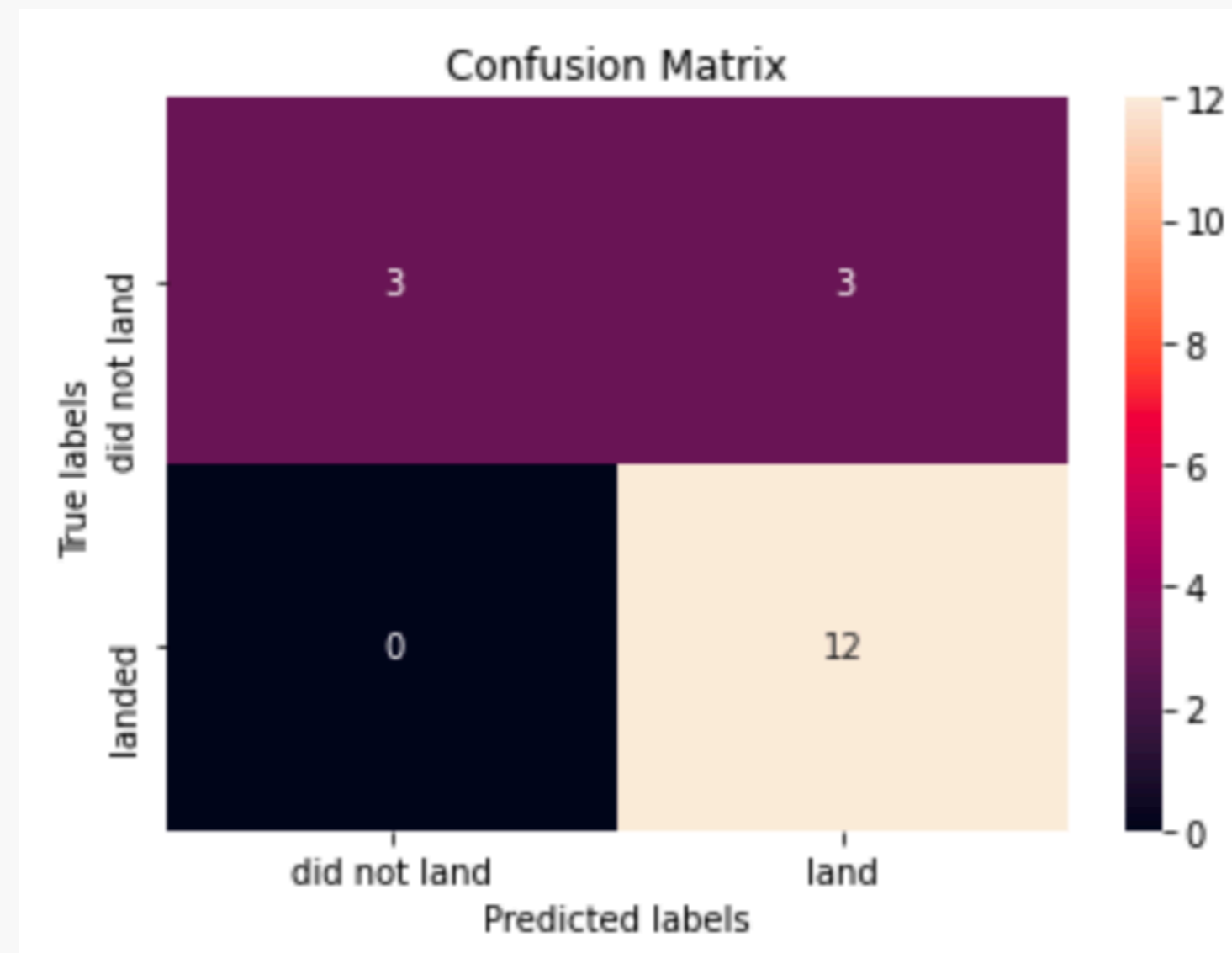
⑤ Predictive Analysis : Classification Models

- K nearest neighbors (KNN)

GridSearchCV best score: 0.8482142857142858

Accuracy score on test set: 0.8333333333333333

Confusion matrix:



RESULTS

⑤ Predictive Analysis : Classification Models

Model selection uses cross validation on the 72 training examples. The decision tree has the highest score in this corrected run:

1. Decision tree: 0.8768
2. KNN: 0.8482
3. Logistic regression: 0.8464
4. SVM: 0.8214

All four models correctly predict 15 of the 18 test examples (83.33%). The small test set does not establish performance on future launches.

DISCUSSION

Feature Impact:

Several variables show meaningful relationships with Falcon 9 landing outcomes. In particular, orbit type, payload mass, and flight number are strongly associated with success. For example, heavier payloads launched into LEO, SSO, and ISS orbits tend to have higher landing success rates, suggesting that certain mission profiles are more favorable for recovery.

GTO Ambiguity:

Geostationary Transfer Orbit (GTO) missions present a more complex pattern. Unlike other orbit types, GTO launches display mixed outcomes across a wide range of payload masses, with both successful and failed landings occurring. This overlap makes GTO cases harder to classify and increases prediction uncertainty for these missions.

Why Machine Learning is Effective:

No single feature is sufficient to fully explain landing success or failure. The outcome depends on interactions between multiple variables rather than isolated effects. Machine learning models are well-suited for this type of problem because they can capture non-linear relationships and complex feature interactions within historical launch data—patterns that would be difficult to identify manually.

Model Performance:

Among the tested models, the Decision Tree achieved the best cross-validation performance. This suggests it is particularly effective at capturing non-linear decision boundaries in the dataset. In contrast, linear models such as Logistic Regression and SVM may struggle to fully represent these complex relationships, while KNN provides competitive but slightly less stable performance.

CONCLUSION

1. This IBM capstone brings together API collection, web scraping, SQL analysis, data preprocessing, interactive visualisation and classification using historical Falcon 9 launch data.
2. The corrected evaluation splits the features before scaling. Each model pipeline fits its scaler within each training fold, and GridSearchCV tunes the models using only the training set.
3. The decision tree is selected using its cross validation score of 0.8768. All four models score 83.33% on the same held out test set, correctly predicting 15 of 18 examples.
4. These results describe an educational experiment using a fixed, preprocessed course dataset. The small test set and random split do not establish reliable forecasting performance for future launches or launch costs.
5. Further evaluation should use more unseen examples and a split that respects launch dates. The original IBM lab attribution is retained, and the updated notebook contains the reproducible evaluation and current outputs.